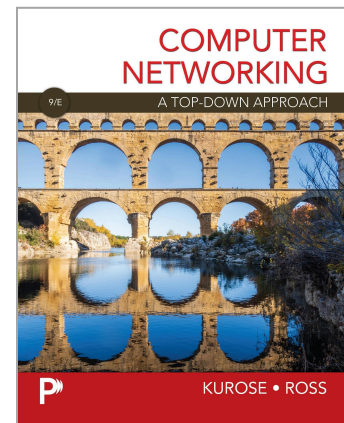


Wireshark Lab: UDP v9

Supplement to *Computer Networking: A Top-Down Approach*, 9th ed., J.F. Kurose and K.W. Ross

“Tell me and I forget. Show me and I remember. Involve me and I understand.” Chinese proverb

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In this lab, we'll take a quick look at the UDP transport protocol. As we saw in Chapter 3 of the text¹, UDP is a streamlined, no-frills protocol. You may want to re-read section 3.3 in the text before doing this lab. Because UDP is simple and sweet, we'll be able to cover it pretty quickly in this lab. So if you've another appointment to run off to in 30 minutes, no need to worry, as you should be able to finish this lab with ample time to spare.

At this stage, you should be a Wireshark expert. Thus, we are not going to spell out the steps as explicitly as in earlier labs. In particular, we are not going to provide example screenshots for all the steps.

The Assignment

Start capturing packets in Wireshark and then do something that will cause your host to send and receive several UDP packets. It's also likely that just by doing nothing (except capturing packets via Wireshark) that some UDP packets sent by others will appear in your trace. In particular, the Domain Name System (DNS) protocol (see section 2.4 in the text; and the DNS Wireshark Lab) typically sends DNS query and response messages inside of UDP, so it's likely that you'll find some DNS messages (and therefore UDP packets) in your trace.

Specifically you can try out the `nslookup` command, which invokes the underlying DNS protocol, which in turn will send UDP segments from/to the host issuing the `nslookup`. `nslookup` is available in most Microsoft, Apple IOS, and Linux operating systems. To run `nslookup` you just type the `nslookup` command on the command line in a DOS window, Mac IOS terminal window, or Linux shell. Figure 1 is a screenshot of running `nslookup` on the Linux command line on the `newworld.cs.umass.edu` host

¹ References to figures and sections are for the 9th edition of our text, *Computer Networks, A Top-down Approach*, 9th ed., J.F. Kurose and K.W. Ross, Addison-Wesley/Pearson, 2025. Our website for this book is http://gaia.cs.umass.edu/kurose_ross You'll find lots of interesting open material there.

located in the CS Department at the University of Massachusetts (UMass) campus, to display the IP address of `www.nyu.edu`.

```
[newworld.cs.umass.edu> nslookup www.nyu.edu
Server:         128.119.240.1
Address:        128.119.240.1#53

Non-authoritative answer:
www.nyu.edu     canonical name = WEB.GSLB.nyu.edu.
Name:   WEB.GSLB.nyu.edu
Address: 216.165.47.12
Name:   WEB.GSLB.nyu.edu
Address: 2607:f600:1002:6113::100
```

Figure 1: the basic `nslookup` command

We don't need to go into any more details about `nslookup` or DNS, as we're just interested in getting a few UDP segments into Wireshark, and we promised this lab would be short!

After starting packet capture on Wireshark, run `nslookup` for a hostname that you haven't visited for a while. Then stop packet capture, set your Wireshark packet filter so that Wireshark only displays the UDP segments sent and received at your host. Pick the first UDP segment and expand the UDP fields in the details window. If you are unable to find UDP segments in your trace or are unable to run Wireshark on a live network connection, you can download a packet trace containing some UDP segments².

² You can download the zip file <http://gaia.cs.umass.edu/wireshark-labs/wireshark-traces-9e.zip> and extract the trace file `dns-wireshark-trace1-1`. This trace file can be used to answer this Wireshark lab without actually capturing packets on your own. This trace was made using Wireshark running on one of the author's computers, while performing the steps indicated in this Wireshark lab. Once you've downloaded a trace file, you can load it into Wireshark and view the trace using the *File* pull down menu, choosing *Open*, and then selecting the trace file name.

Answer the following questions. Submit your answers as a pdf on Canvas.

1. Select the first UDP segment in your trace.
 - a. What is the packet number³ of this segment in the trace file?

1

- b. What type of application-layer payload or protocol message is being carried in this UDP segment? Look at the details of this packet in Wireshark.

DNS

- c. How many fields are there in the UDP header? (You shouldn't look in the textbook! Answer these questions directly from what you observe in the packet trace.) What are the names of these fields?

4 fields: src port, dst port, length, checksum

2. By consulting the displayed information in Wireshark's packet content field for this packet (or by consulting the textbook), what is the length (in bytes) of each of the UDP header fields? (You can click on the field and it should show the bytes at the very bottom)

2 bytes each

3. The value in the Length field is the length of what? (You can consult the text for this answer). Verify your claim with your captured UDP packet.

The value in the length field is of the UDP header + the data. Since the length value is 50, and the UDP header is 8 bytes (2 for each field) then there are 42 bytes for the data.

4. What is the maximum number of bytes that can be included in a UDP payload? (Hint: the answer to this question can be determined by your answer to 2. above)

65,527 bytes

5. What is the largest possible source port number? (Hint: see the hint in 4.)

65,535 bytes

6. Examine the pair of UDP packets in which your host sends the first UDP packet and the second UDP packet is a reply to this first UDP packet. (Hint: for a second packet to be sent in response to a first packet, the sender of the first packet should be the destination of the second packet).

³ Remember that this "packet number" is assigned by Wireshark for listing purposes only; it is NOT a packet number contained in any real packet header.

- a. What is the packet number⁴ of the first of these two UDP segments in the trace file?

1

- b. What is the value in the source port field and destination port field in this UDP segment?

Src: 56214

Dst: 53

- c. What is the packet number⁵ of the second of these two UDP segments in the trace file?

2

- d. What is the value in the source port field and destination port field in this second UDP segment?

Src: 53

Dst: 56214

- e. Describe the relationship between the port numbers in the two packets.

The port numbers are reversed. The request sends data from the client's ephemeral port (56214) to the server's DNS port (53). The reply sends data from the server's DNS port (53) back to the client's ephemeral port (56214) to ensure the message reaches the correct application process.

That's it! As a streamlined, no-frills protocol , UDP deserves a streamlined, no-frills Wireshark Lab ☺ .

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